

NexusAI - AI/ML Engineer Job Description

Job Title: AI/ML Engineer - Retrieval & LLM Systems

Company: NexusAI Intelligence

Location: Austin, TX / Hybrid

Role Overview:

NexusAI is seeking an ambitious AI/ML Engineer to build state-of-the-art enterprise Retrieval-Augmented Generation (RAG) pipelines, fine-tuned Large Language Models, and real-time inference services.

Key Responsibilities:

- Build and optimize end-to-end RAG pipelines with vector databases (Pinecone, Milvus, Chroma) and hybrid semantic search.
- Fine-tune open-source Large Language Models (LLMs) like LLaMA and Mistral using LoRA and QLoRA.
- Deploy scalable ML models and APIs using FastAPI, Docker, and Kubernetes on AWS/GCP cloud platforms.
- Architect automated data pipelines using Apache Airflow and PySpark for processing unstructured text at scale.
- Implement comprehensive evaluation frameworks (BLEU, ROUGE, RAGAS) to benchmark hallucination and retrieval accuracy.
- Collaborate with software engineers to integrate AI microservices into enterprise production environments.

Requirements:

- Master's or Bachelor's degree in Data Science, Computer Science, or related quantitative field.
- Strong proficiency in Python, PyTorch, Scikit-learn, and SQL.
- Practical experience building RAG systems, LangChain/LlamaIndex agents, and vector embeddings.
- Experience with cloud platforms (AWS, GCP) and containerization with Docker.
- Solid understanding of statistical testing, model evaluation metrics, and natural language processing.

Bonus Points:

- Experience with PySpark or distributed data processing.
- Familiarity with CI/CD for machine learning (MLOps, MLflow, Weights & Biases).
- Contributions to open-source LLM or RAG projects.